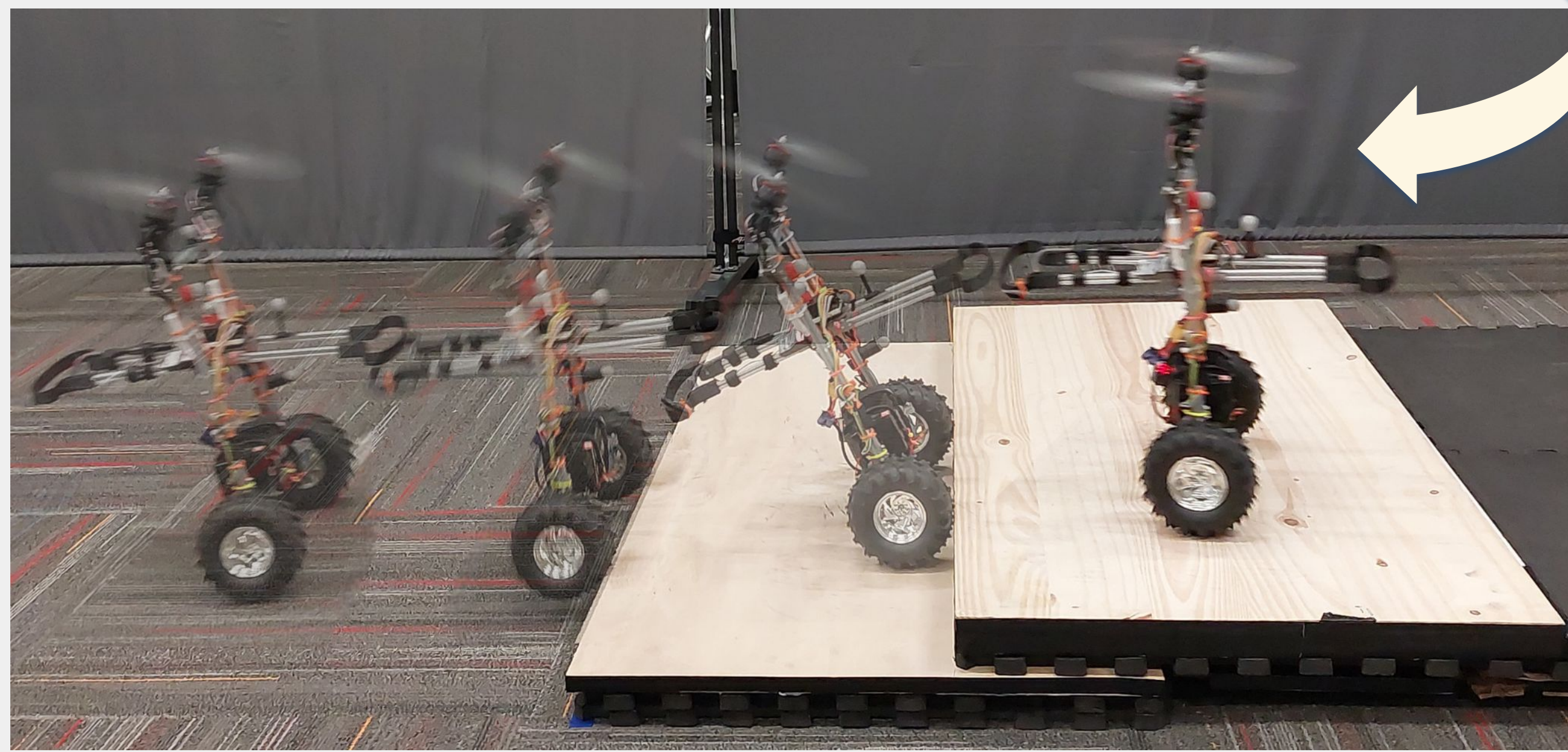


Motivation

Hybrid aerial-ground robots face a fundamental trade-off on stair-like terrain: **wheels alone stall** at step edges, while **full flight wastes energy** for small height gains. DoubleBee's¹ tilting propellers provide additional control authority during ground contact, **enabling thrust-assisted traversal** without mode-switching overhead.

Q. Can energy-efficient hybrid actuation emerge from learning alone, without prescribing discrete locomotion modes?



Thrust-assisted climbing on DoubleBee. Four exposures of one run over two 6 cm steps, 40 cm apart. The wheels drive forward while thrust carries each edge, covering 1.3 m in 2.9 s before settling upright on the top step.

Methodology**1. Platform & Modelling:**

- 2 tilting propellers + 2 active wheels, 3.22 kg, 6-DOF
- Low-level control commands at time t :

$$u_t = [u_{p,1,t}^{pwm}, u_{p,2,t}^{pwm}, \omega_{w,1,t}^{ref}, \omega_{w,2,t}^{ref}, \sigma_{1,t}^{ref}, \sigma_{2,t}^{ref}]^\top$$

2. Sim Setup:

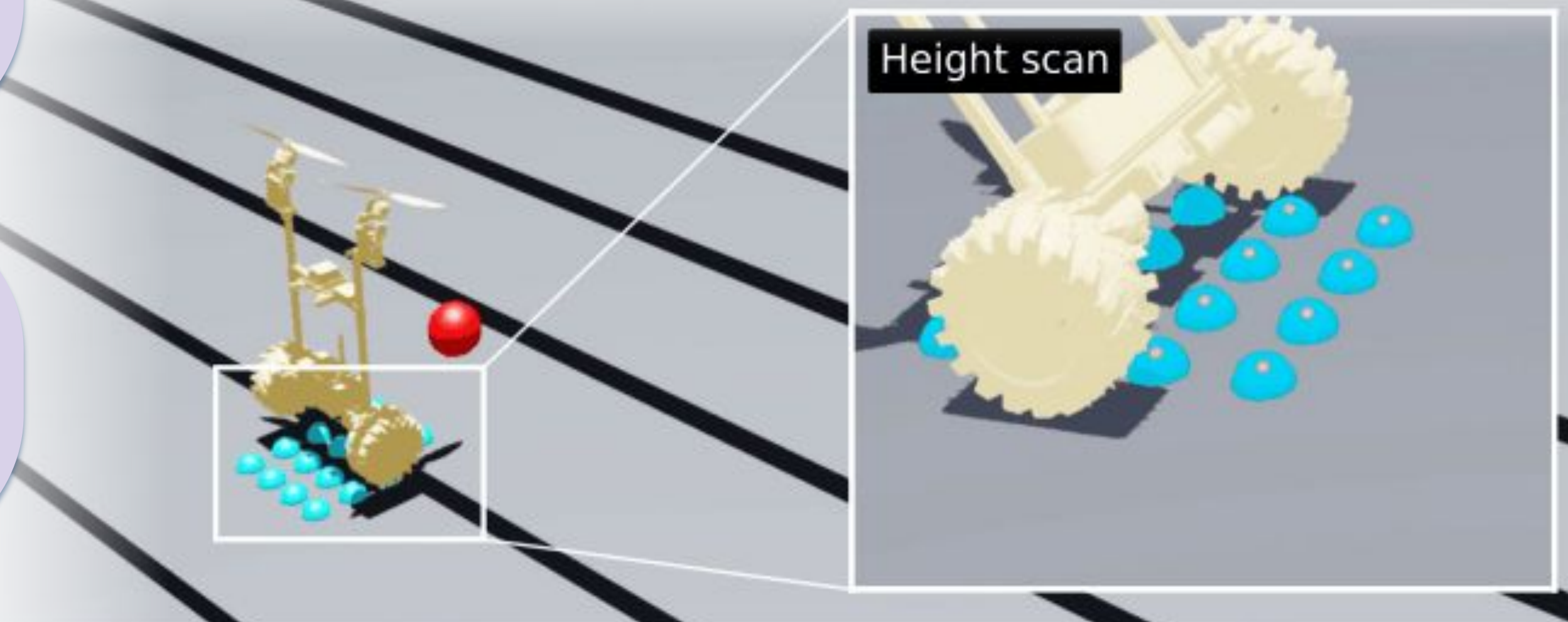
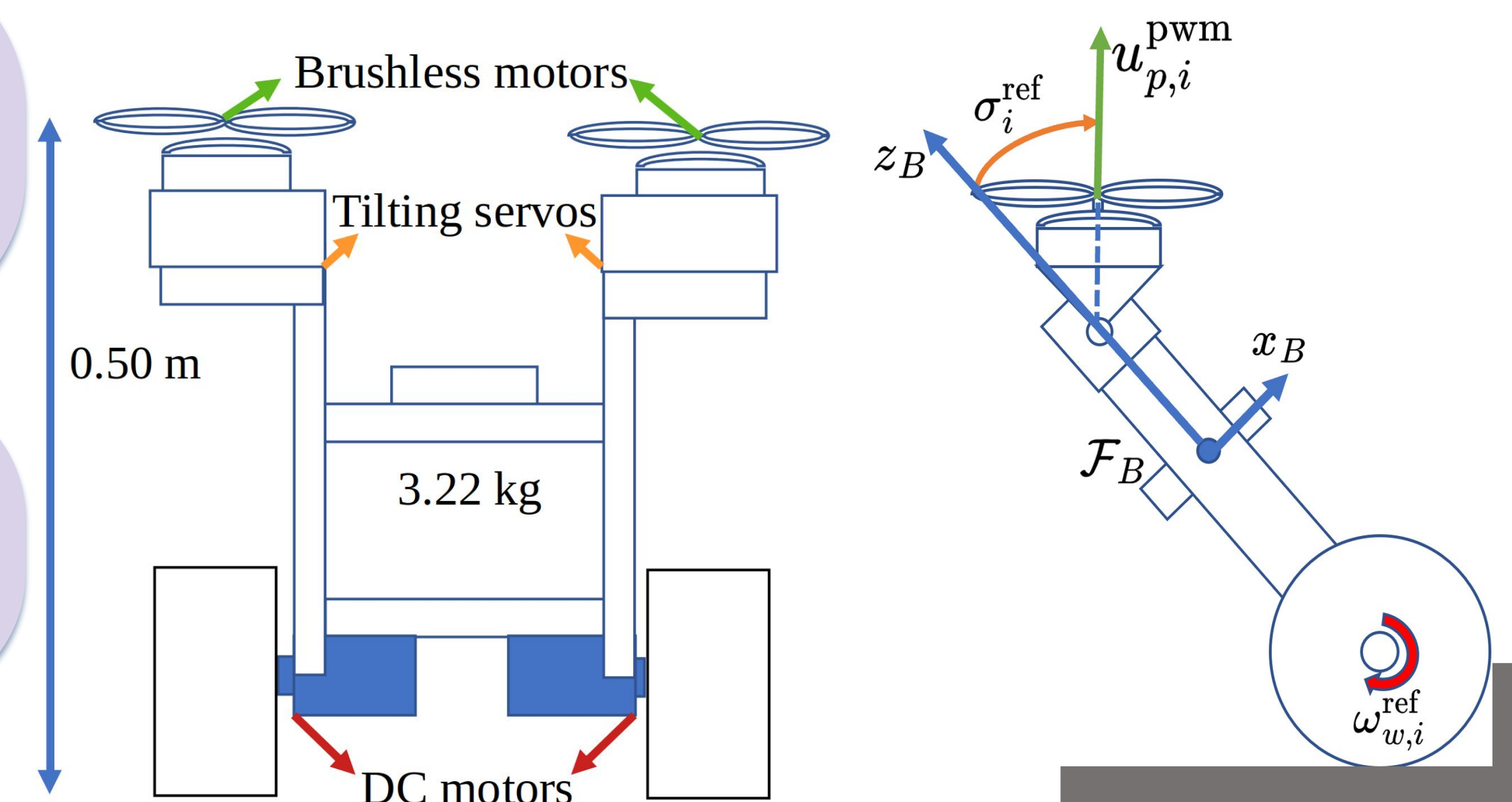
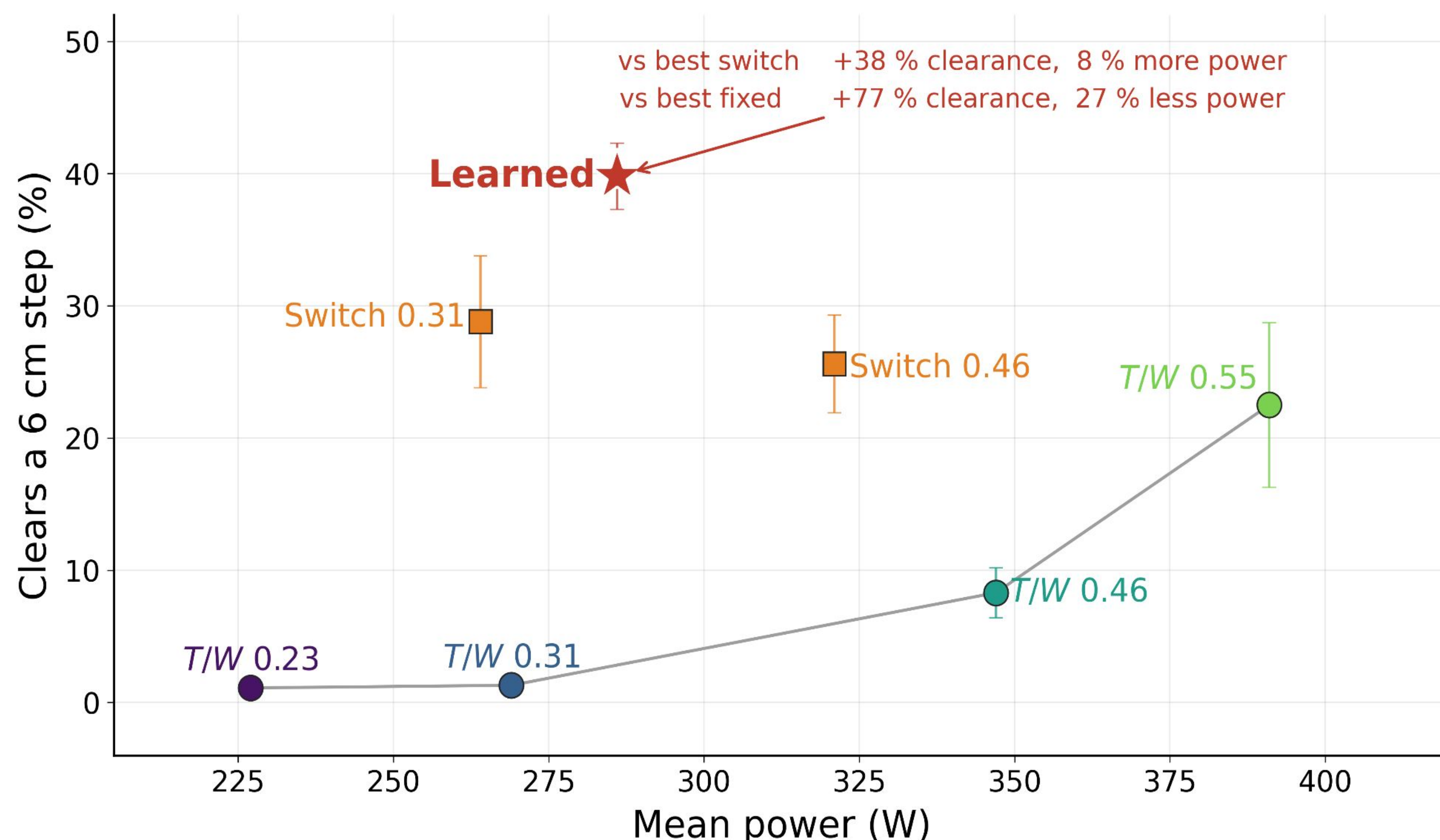
- Isaac Lab, 1024 parallel envs, **3→8 cm curriculum**
- 25 procedural staircase tiles, 0.40 m tread
- SysId** 4th-order PWM→thrust polynomial

3. RL Training:

- TQC off-policy · $o_t \in \mathbb{R}^{41}$ · $a_t \in [-1, 1]^4$
- $r_t^E = \exp(-E_t/E_{ref}) - 1$, **penalises electrical energy**
- Goal gated on distance, height, posture and body rate

4. Sim2Real

- Randomised per reset: mass $\pm 5\%$, thrust $\pm 20\%$, friction, damping, pushes
- Observations rebuilt at 50 Hz from mocap, encoders and terrain map

**Results**

In simulation, wheels alone never clear a 6 cm step, while flight alone costs about **2.5 times the power** the learned policy uses to climb, motivating **hybrid actuation**. We compare against four fixed thrust levels and two switching controllers that raise thrust on detecting a step. At 6 cm the learned policy clears **1.8 times as many steps** as the best fixed level while drawing 286 W against 391 W, and it leads both switches by 10 to 15 points from 5 cm to 7 cm. Both switches reach the same peak thrust as the policy, so the improvement stems from **continuous thrust modulation** rather than from the magnitude available.

On hardware, the policy clears both 6 cm steps in **8 of 10 trials** at 275 W, close to the 286 W predicted in simulation. We further test robustness on **three additional terrains**, closely spaced unequal steps, two ramps meeting at a crest, and taller steps topped with foam. The policy lifts the wheels onto 8 and 9 cm steps, clearing heights above the wheel radius and beyond the trained geometry. It traverses the ramp while the supplied height scan still described the primary staircase, pointing to a useful **role for proprioceptive feedback**.

Future Work

Three gaps stand out. The policy never backs off and retries after a failed approach, so teaching an explicit **recovery behaviour** would stop a stalled attempt sitting against the step and burning energy without traversing. **Modelling wheel contact** across terrain materials, rather than one rigid surface, would address the slip that simulation never reproduced. **Moving perception onboard** is the remaining step towards sustained multi-step climbing, and since stable climbs are rare early in training, **seeding with priors** distilled from the policy's own successful climbs may get there faster than reward shaping alone.

References

¹M. Cao, X. Xu, S. Yuan, K. Cao, K. Liu, and L. Xie, "DoubleBee: A Hybrid Aerial-Ground Robot with Two Active Wheels", Mar 2023, arXiv:2303.05075 [cs.RO]. Available: <https://arxiv.org/abs/2303.05075>

Acknowledgments

This work is supported by the German Academic Exchange Service (DAAD) under the RISE Weltweit Program. Thank you to Dr. Muqing Cao and Dr. Sebastian Scherer for their mentorship, Dr. Dorit Borrmann for her continued support, and Rachel Burcin and Dr. John Dolan for coordinating the RISS program.